

Estimating Patient Independence with Sleep Sensors

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ABSTRACT

The aging population problem is growing worldwide and physical therapists must be able to care for the elderly and injured efficiently. One burden facing physical therapists is determining the Functional Independence Measure (FIM), which measures the level of independence in activities of daily living for the elderly and injured. This measure is used in designing rehabilitation programs. Determining the FIM is a time-consuming task for both physical therapists and rehabilitation patients because it requires an interview. Some researchers have explored estimating FIM using wearable devices; however, it is burdensome for patients to wear such devices continuously. Therefore, we propose a method to estimate FIM motor items automatically using machine learning and sleep sensors. The proposed method classifies patients into three levels of FIM values that are referred to as s-FIM using vital data acquired through sleep sensors, personal data (age, gender, and Body Mass Index), and s-FIM values from the previous day. The results of a one-week study based on 19 patients showed that the proposed method had a mean accuracy of 0.87 for s-FIM motor items. The results were more accurate than continuing to use the s-FIM values determined by the physical therapist without updating them for one week.

CCS CONCEPTS

• Human-centered computing → Empirical studies in ubiquitous and mobile computing.

KEYWORDS

Functional independence measure, Machine learning, Sleep data

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1 INTRODUCTION

The ratio of the elderly in the general population that require care is increasing due to the aging problem. Accordingly, there is a need for technology to assist physical therapists in providing more efficient care in their role of aiding the elderly and injured in rehabilitation. During the rehabilitation process, physical therapists determine the Functional Independence Measure (FIM), which represents the level of independence of daily activities of a person requiring care, and use this index to design training programs. The FIM consists of 13 items related to motor functions, e.g., transfer ability, and 5 items related to cognitive functions, e.g., communications. Each item is divided into seven levels. Although, determining the FIM is an important task to assess the independence level of the patient, it is time-consuming for both the patient and the physical therapist because it requires an interview. Therefore, supporting physical therapists by automating this time-consuming determination process for FIM is an important issue.

Researchers have been exploring methods to measure FIM automatically by analyzing data from wearable sensors. Conci *et al.* [1] used commercially available wearable sensors to measure the patient physical activity, heart rate, and sleep quality during inpatient rehabilitation and to predict the FIM at discharge. Oishi *et al.* [4] applied a machine learning approach to data collected from a variety of sensors to predict the FIM including wearable sensors, e.g., smart watches and RFID-equipped clothing; and environmental sensors, e.g., temperature sensors and illumination sensors. These studies using wearable sensors are expected to yield predictions with a high degree of accuracy because these devices can constantly monitor the patient vital data. However, there are hurdles to using this approach in hospitals because it is burdensome for patients undergoing rehabilitation to wear the device constantly and to ensure that the wearable devices are charged.

Thus, we propose estimating the degree of patient independence by simply placing a sleep sensor under the bed. Since this method can be operated by determining the initial FIM value and simply placing a sleep sensor under the bed, we consider that it is more practical and less burdensome for patients than using wearable sensors. We select 13 FIM motor items as the target for estimation considering medical findings based on a relationship between sleep and rehabilitation outcomes [2]. In addition, we design the estimated FIM as a three level score, which we refer to as the s-FIM (full assistance, partial assistance, and independence) based on interviews with physical therapists.

Proposed Method: Figure 1 shows the flow of estimating an item of s-FIM. We apply this flow to the 13 items in s-FIM. There are three types of data used in this estimation: sleep sensor, personal data, and the FIM value from the previous day (FIM-pd). The time series data acquired from the sleep sensors are the respiratory rate, heart rate,

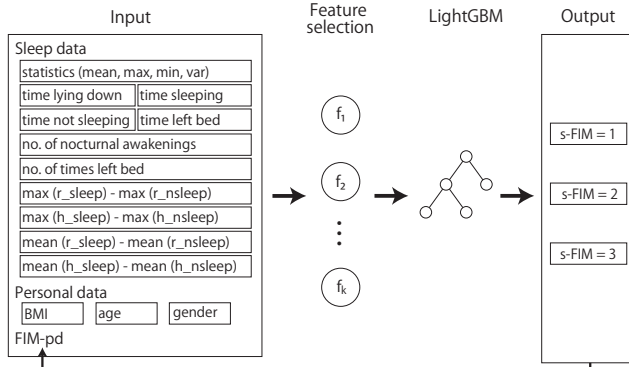


Figure 1: Outline of s-FIM estimation method

activity, respiratory event index, periodic limb movement disorder index, and sleep detection. These data are processed as shown in Fig. 1 based on the expertise of a physical therapist. The statistics are calculated as the mean, maximum, minimum, and variance of the five types of sleep sensors (excluding sleep detection), separated by three hours. The abbreviations “h_sleep” and “r_sleep” are used to represent heart rate and respiration rate during sleep, and the suffix “nsleep” is used to represent non-sleep. The personal data consists of age, gender, and Body Mass Index. The FIM-pd is a parameter comprising the s-FIM value estimated using the model on the previous day. In other words, the FIM-pd requires a physical therapist assessment value on only the first day of the experiment. Taking the above data as input, we select the top k features of feature importance and applied LightGBM [3] to estimate the s-FIM.

2 EVALUATION

The subjects were recruited from patients in the acute care ward of a rehabilitation hospital. The hospital institutional review board approved the study and all participants gave written informed consent before they participated. There were 19 patients (5 males and 14 females, 66–96 years of age) who participated in the study during hospitalization. The patient data obtained over one week consists of personal data, sleep data, and FIM assessed by a physical therapist. There were cases in which data failed to be collected for one day in six among the 19 patients. The sleep data collected through the sleep sensor, which is a Nemuri SCAN (Model NN-1520, PARAMOUNT BED Co., Ltd.), included time-series data in minutes and the interval to be analyzed per day was from 12:00 p.m. to 12:00 p.m. In this study, we estimated s-FIM motor items using LightGBM trained on a leave-one-subject-out cross-validation basis. Finally, we compared the accuracy of the proposed method to that of the baseline where the first s-FIM value measured by the physical therapist was used as the daily estimated result. This accuracy is simply the rate of correct classifications.

The results show that the mean accuracy of the s-FIM motor items is 0.80 (standard deviation: 0.09) for the baseline and 0.87 (standard deviation: 0.04) for the proposed method, and the accuracy of the proposed method exceeds that of the baseline. Furthermore, as shown in Fig. 2, the estimation accuracies are different for each item, with the highest accuracy of 0.94 for the eating item and the lowest accuracy of 0.79 for the bladder management item. The

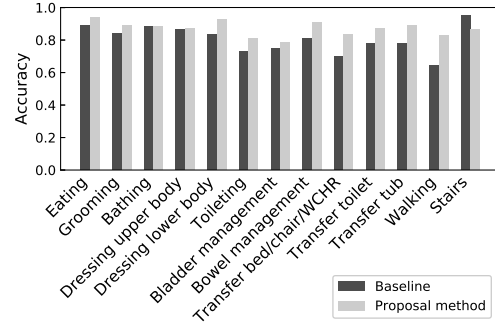


Figure 2: Accuracy of baseline and proposed method for each s-FIM motor item

accuracy of the proposed method is higher than that for the baseline for all items except for the stairs item. These results show that the proposed method tracks the patient level of independence as it changes during hospitalization. We consider that the lower accuracy of the proposed method than that for the baseline for the stairs item is due to underfitting by imbalanced data (86 samples for s-FIM = 1, likewise 15 samples for 2 and 26 samples for 3). Therefore, we believe that appropriate learning for such imbalanced data can be achieved by upsampling or increasing the amount of training data.

We find that through this experiment some features related to heart rate are of high feature importance for estimating s-FIM. For example, for the toileting item, the top three feature importance items are the FIM-pd, variance of respiratory rate from 3 a.m. to 6 a.m., and the maximum heart rate from 9 a.m. to 12 a.m. Prior to this experiment, we expected that the heart rate would account for a high percentage of the features involved in estimating the s-FIM, since a physical therapist had indicated that there is a relationship between motor function and cardiac function. This suggests that patients with weakened motor function may have their respiratory function affected due to chest compression in the supine position.

3 CONCLUSION

We proposed a method that estimates the s-FIM with sleep sensors. The results show that the mean accuracy of estimating the s-FIM motor items is 0.87. The study results are expected to be applied in hospitals because the proposed method can estimate the s-FIM without burdening patients and physical therapists. We focused on the motor items of the s-FIM for estimation and evaluation, although we will try to estimate the cognitive items in the future.

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